

ERKKA KIUKKONEN

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SUMMARY

Computer Science student at Aalto University (121/180 ECTS). Proficient in TypeScript, JavaScript, Node.js, React, Rust, and C/C++. Hands-on experience in research software, distributed systems, and hardware. Strong communicator with leadership experience across student organisations and military service. Open to internship and part-time software engineering opportunities.

TECHNICAL SKILLS

Languages TypeScript, JavaScript, Rust, C/C++, Scala, Python
Frontend React, SvelteKit, WebGL, WASM, HTML5/CSS3
Backend & Systems Node.js, WebSocket, REST APIs, distributed systems
Hardware Arduino, KiCad (PCB design)
Tools Git, GitHub, Linux, Microsoft 365

PROJECT

Regenie Nonlinear · FIMM, University of Helsinki | 2025 — ongoing

- Forked REGENIE 4.1.2 to add nonlinear phenotype modelling for biobank-scale GWAS; preserves the original memory-efficient block-ridge design.
- Deep performance work on a large C/C++ codebase — every added allocation visible in benchmarks at biobank scale. Open source: github.com/Erkkuleo/regenie-nonlinear

WORK EXPERIENCE

Research Trainee · FIMM, University of Helsinki | Aug 2025 — present

- Contributing to bioinformatics research on circadian rhythms using large-scale genomic data; extended the REGENIE toolkit with nonlinear phenotype modelling (C/C++).

Arduino Coding Teacher · Etelä-Tapiola Upper Secondary School | 2021 – 2023

- Designed and delivered hands-on Arduino programming curriculum; introduced students to embedded systems, sensor programming, and iterative prototyping.

Logistics Worker · Kesko | May – Aug 2023

- Warehouse operations, inventory management, and time-critical logistics.

EDUCATION

B.Sc. Computer Science · Aalto University | 2023 — present

- 102 / 180 ECTS completed. Focus: distributed systems, algorithms, software engineering.

Second Lieutenant · Reserve Officer School, Course 265 | 2024 – 2025

- Officer training and leadership qualification, Finnish Defence Forces.

Finnish Matriculation Examination · Etelä-Tapiola Upper Secondary School | 2023

VOLUNTEERING & LEADERSHIP

Head of Corporate Relations · Tietokilta ry (Aalto CS Guild) | 2026

- Leading partnerships, company events, and sponsorship strategy for one of Finland's largest technology student guilds.

Board Member / Chair · Uudenmaan partiopiiri · Veikkolan Eräveikot ja Haltiat ry | 2023 – 2024

- Strategic governance for regional Scout organisations; chaired local group board in 2023.

LANGUAGES

Finnish Native
English Excellent
Swedish Good
French Good